StageIV Model - Analysis Report 24 June 2026, 16:25

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Analysis Parameters

Parameter	Analysis	Value
modelReference	all	Recon3D
objFunction	all	biomass_reaction
osenseStr	FBA	max
minNorm	FBA	zero
optPercentage	FVA	90
osenseStr	sampling	max
minNorm	sampling	zero
objThreshold	sampling	0.9
sampleFile	sampling	sampleFile
samplerName	sampling	CHRR
loopless	loopless	1
samplingToUse	loopless	
options.toRound	sampling	1
options.countSampleProcesses	sampling	5
options.nPointsReturned	sampling	5000
options.nFiles	sampling	10
Options.nWorkers	sampling	5
options.numCores	sampling	10
options.maxTime	sampling	36000
options.nWarmupPoints	sampling	10*size(model.S, 2)
options.nStepsPerPoint	sampling	round(size(model.S, 2))
modelSampling	sampling	model
path	sampling	samplingResults/CHRR/
nPointsReturned	kld	1000
numberOfIndSamplings	kld	8
objThreshold	kld	0.9
method	singleGeneDeletion	FBA
geneList	singleGeneDeletion	{'26.1', '314.1'}
method	doubleGeneDeletion	FBA
geneList1	doubleGeneDeletion	
geneList2	doubleGeneDeletion	

Model Characteristics

model Consistency	nb Rxns	nb Consistent Rxns	nb Uptake Rxns	nb GPRrules
YES	2519	2519	35	2519

Medium Composition

Mets_Recon3D	ExRxns_Recon3D	Concentration_uM
4hpro_LT[e]	EX_4hpro[e]	1525.2
Lcystin[e]	EX_Lcystin[e]	2075.2199999999998
aqcobal[e]	EX_aqcobal[e]	0.036890300000000001
arg_L[e]	EX_arg_L[e]	11481.099999999999
asn_L[e]	EX_asn_L[e]	3784.4400000000001
asp_L[e]	EX_asp_L[e]	1502.52
btn[e]	EX_btn[e]	8.1863200000000003
ca2[e]	EX_ca2[e]	4239.8400000000001
chol[e]	EX_chol[e]	214.869
cl[e]	EX_cl[e]	1087720
fol[e]	EX_fol[e]	22.655199999999997
glc_D[e]	EX_glc_D[e]	111012
gln_L[e]	EX_gln_L[e]	20528.299999999999
glu_L[e]	EX_glu_L[e]	1359.3399999999999
gly[e]	EX_gly[e]	1332.0899999999999
gthrd[e]	EX_gthrd[e]	32.539400000000001
hco3[e]	EX_hco3[e]	238067
his_L[e]	EX_his_L[e]	966.80600000000004
ile_L[e]	EX_ile_L[e]	3811.8499999999999
inost[e]	EX_inost[e]	1942.72
k[e]	EX_k[e]	53655.300000000003
leu_L[e]	EX_leu_L[e]	3811.8499999999999
lys_L[e]	EX_lys_L[e]	2189.98
met_L[e]	EX_met_L[e]	1005.29
na1[e]	EX_na1[e]	1380530
ncam[e]	EX_ncam[e]	81.886699999999999
phe_L[e]	EX_phe_L[e]	908.04499999999996
pi[e]	EX_pi[e]	56353.899999999994
pnto_R[e]	EX_pnto_R[e]	10.4923
pro_L[e]	EX_pro_L[e]	1737.1699999999998
pydxn[e]	EX_pydxn[e]	48.628700000000002
ribflv[e]	EX_ribflv[e]	5.3140599999999996
ser_L[e]	EX_ser_L[e]	2854.7000000000003
so4[e]	EX_so4[e]	4057.8299999999999
thm[e]	EX_thm[e]	29.650699999999997
thr_L[e]	EX_thr_L[e]	1678.98
trp_L[e]	EX_trp_L[e]	244.822
tyr_L[e]	EX_tyr_L[e]	1110.3
val_L[e]	EX_val_L[e]	1707.21

Table of Exchangers

FBA for objective function: 45.0528

ReactionID	FBA	FVAmin	FVAmax	NormalizedFBA	NormalizedFVAmin	NormalizedFVAmax
EX_o2[e]	-999.7750	-1000.0000	-414.3254	-22.1912	-22.1962	-9.1964
EX_gln_L[e]	-549.2802	-978.4781	-187.6922	-12.1919	-21.7185	-4.1660
EX_glc_D[e]	-521.0941	-972.1950	-451.2757	-11.5663	-21.5790	-10.0166
EX_arg_L[e]	-138.5227	-337.2860	-14.5671	-3.0747	-7.4865	-0.3233
EX_pi[e]	-43.9807	-144.2247	-39.5826	-0.9762	-3.2012	-0.8786
EX_gthrd[e]	-32.5394	-32.5394	-22.9249	-0.7222	-0.7222	-0.5088
EX_h2o[e]	-31.5662	-469.0083	963.7886	-0.7006	-10.4102	21.3924
EX_lys_L[e]	-26.6764	-211.7031	-24.0088	-0.5921	-4.6990	-0.5329
EX_leu_L[e]	-24.5783	-963.5822	-22.1205	-0.5455	-21.3878	-0.4910
EX_tyr_L[e]	-21.0535	-331.3662	-11.1378	-0.4673	-7.3551	-0.2472
EX_ser_L[e]	-18.5368	-71.5666	-16.6831	-0.4114	-1.5885	-0.3703
EX_val_L[e]	-15.8859	-321.1586	-14.2973	-0.3526	-7.1285	-0.3173
EX_thr_L[e]	-14.0876	-736.1071	-12.6788	-0.3127	-16.3388	-0.2814
EX_ile_L[e]	-12.8886	-318.4610	-11.5998	-0.2861	-7.0686	-0.2575
EX_asn_L[e]	-12.5889	-12.6470	-11.3300	-0.2794	-0.2807	-0.2515
EX_phe_L[e]	-11.6897	-485.7550	-10.5207	-0.2595	-10.7819	-0.2335
EX_pchol_hs[e]	-11.2920	-11.3442	-10.1628	-0.2506	-0.2518	-0.2256
EX_met_L[e]	-6.8939	-6.9258	-6.2045	-0.1530	-0.1537	-0.1377
EX_his_L[e]	-5.6949	-340.2032	-5.1255	-0.1264	-7.5512	-0.1138
EX_chol[e]	-4.7189	-101.1928	-1.4729	-0.1047	-2.2461	-0.0327
EX_inost[e]	-1.0504	-1.0553	134.2174	-0.0233	-0.0234	2.9791
EX_crm_hs[e]	-0.7878	-0.7914	-0.7090	-0.0175	-0.0176	-0.0157
EX_pglyc_hs[e]	-0.6565	-0.6595	-0.5909	-0.0146	-0.0146	-0.0131
EX_trp_L[e]	-0.5995	-244.8220	-0.5395	-0.0133	-5.4341	-0.0120
EX_2hb[e]	0.0000	0.0000	723.4283	0.0000	0.0000	16.0573
EX_34dhoxpeg[e]	0.0000	0.0000	246.0992	0.0000	0.0000	5.4625
EX_3mlda[e]	0.0000	0.0000	190.0339	0.0000	0.0000	4.2180
EX_ala_B[e]	0.0000	0.0000	31.0719	0.0000	0.0000	0.6897
EX_ala_D[e]	0.0000	0.0000	607.2141	0.0000	0.0000	13.4778
EX_bilglcur[e]	0.0000	0.0000	1.1674	0.0000	0.0000	0.0259
EX_chsterol[e]	0.0000	0.0000	5.5395	0.0000	0.0000	0.1230
EX_co[e]	0.0000	0.0000	1.1674	0.0000	0.0000	0.0259
EX_cspg_b[e]	0.0000	0.0000	4.0589	0.0000	0.0000	0.0901
EX_gthox[e]	0.0000	0.0000	4.8073	0.0000	0.0000	0.1067
EX_mercplaccys[e]	0.0000	0.0000	15.3255	0.0000	0.0000	0.3402
EX_ncam[e]	0.0000	-81.8867	0.0000	0.0000	-1.8176	0.0000
EX_oxa[e]	0.0000	0.0000	8.0096	0.0000	0.0000	0.1778
EX_sl_L[e]	0.0000	0.0000	30.6511	0.0000	0.0000	0.6803
EX_thym[e]	0.0000	0.0000	2.9804	0.0000	0.0000	0.0662
EX_tsul[e]	0.0000	0.0000	30.6511	0.0000	0.0000	0.6803
EX_xoltri27[e]	0.0000	0.0000	5.5056	0.0000	0.0000	0.1222
EX_prpp[e]	0.0000	0.0000	19.0238	0.0000	0.0000	0.4223
EX_pydx5p[e]	0.0000	0.0000	48.6287	0.0000	0.0000	1.0794
EX_so3[e]	0.0000	-46.5484	30.6511	0.0000	-1.0332	0.6803

ReactionID	FBA	FVAmin	FVAmax	NormalizedFBA	NormalizedFVAmin	NormalizedFVAmax
EX_CE0074[e]	0.0000	0.0000	8.3861	0.0000	0.0000	0.1861
EX_4hpro[e]	0.0000	-318.9484	169.0792	0.0000	-7.0794	3.7529
EX_leuleu[e]	0.0000	0.0000	470.7309	0.0000	0.0000	10.4484
EX_5oxpro[e]	0.0000	0.0000	135.1627	0.0000	0.0000	3.0001
EX_idour[e]	0.0000	0.0000	3.3847	0.0000	0.0000	0.0751
EX_cala[e]	0.0000	0.0000	3.0545	0.0000	0.0000	0.0678
EX_3hmp[e]	0.0000	0.0000	212.7166	0.0000	0.0000	4.7215
EX_4aabutn[e]	0.0000	0.0000	193.0896	0.0000	0.0000	4.2858
EX_3moxtyr[e]	0.0000	0.0000	204.8091	0.0000	0.0000	4.5460
EX_glyclt[e]	0.0000	0.0000	318.9484	0.0000	0.0000	7.0794
EX_pac[e]	0.0000	0.0000	294.8684	0.0000	0.0000	6.5449
EX_1mncam[e]	0.0000	0.0000	81.8867	0.0000	0.0000	1.8176
EX_progly[e]	0.0000	0.0000	9.5829	0.0000	0.0000	0.2127
EX_thbpt[e]	0.0000	0.0000	7.9291	0.0000	0.0000	0.1760
EX_argglygly[e]	0.0000	0.0000	4.8073	0.0000	0.0000	0.1067
EX_trpglyleu[e]	0.0000	0.0000	9.6145	0.0000	0.0000	0.2134
EX_homoval[e]	0.0000	0.0000	186.4611	0.0000	0.0000	4.1387
EX_glypro[e]	0.0000	0.0000	9.5829	0.0000	0.0000	0.2127
EX_ppi[e]	0.0000	0.0000	43.3812	0.0000	0.0000	0.9629
EX_so4[e]	0.0000	-16.0193	0.0000	0.0000	-0.3556	0.0000
EX_urea[e]	0.0000	0.0000	8.3861	0.0000	0.0000	0.1861
EX_pnto_R[e]	0.0000	-7.2508	0.0000	0.0000	-0.1609	0.0000
EX_ind3ac[e]	0.0000	0.0000	244.2825	0.0000	0.0000	5.4221
EX_ac[e]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
EX_etha[e]	0.0000	0.0000	102.1382	0.0000	0.0000	2.2671
EX_galt[e]	0.0000	0.0000	67.5814	0.0000	0.0000	1.5000
EX_glcr[e]	0.0000	0.0000	21.7045	0.0000	0.0000	0.4818
EX_ptrc[e]	0.0000	0.0000	174.8682	0.0000	0.0000	3.8814
EX_pydxn[e]	0.0000	-48.6287	0.0000	0.0000	-1.0794	0.0000
EX_phpyr[e]	0.0000	0.0000	475.2343	0.0000	0.0000	10.5484
EX_2h3mv[e]	0.0000	0.0000	306.8612	0.0000	0.0000	6.8111
EX_2hiv[e]	0.0000	0.0000	306.8612	0.0000	0.0000	6.8111
EX_2m3hbu[e]	0.0000	0.0000	150.1612	0.0000	0.0000	3.3330
EX_C05769[e]	0.0000	0.0000	1.1791	0.0000	0.0000	0.0262
EX_im4ac[e]	0.0000	0.0000	335.0777	0.0000	0.0000	7.4374
EX_dopa4glcur[e]	0.0000	0.0000	40.3471	0.0000	0.0000	0.8956
EX_dopa3glcur[e]	0.0000	0.0000	40.3471	0.0000	0.0000	0.8956
EX_34dhpe[c]	0.0000	0.0000	288.5113	0.0000	0.0000	6.4038
EX_M00117[e]	0.0000	0.0000	16.6868	0.0000	0.0000	0.3704
EX_dolichol_L[e]	0.0000	0.0000	0.1548	0.0000	0.0000	0.0034
EX_q10h2[e]	5.1818	4.6636	7.8971	0.1150	0.1035	0.1753
EX_ala_L[e]	21.0535	0.0000	607.2141	0.4673	0.0000	13.4778
EX_lac_D[e]	65.9900	0.0000	230.0747	1.4647	0.0000	5.1068
EX_argalaala[e]	122.3371	0.0000	320.0266	2.7154	0.0000	7.1034

ReactionID	FBA	FVAmin	FVAmax	NormalizedFBA	NormalizedFVAmin	NormalizedFVAmax
EX_glu_L[e]	230.2673	1.0749	774.6297	5.1111	0.0239	17.1938
EX_nh4[e]	521.6134	156.4238	1000.0000	11.5778	3.4720	22.1962
EX_h[e]	526.9692	-723.3958	1000.0000	11.6967	-16.0566	22.1962
EX_lac_L[e]	728.5236	89.3018	1000.0000	16.1704	1.9822	22.1962
EX_hco3[e]	994.8182	609.1572	1000.0000	22.0811	13.5210	22.1962

Fluxes Per Pathway

Citric acid cycle

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVAmin	Normali-zedFVA-max
ACITL	atp[c] + coa[c] + cit[c] -> adp[c] + pi[c] + accoa[c] + oaa[c]	0.0000	0.0000	144.3215	0.0000	0.0000	3.2034
ACONTm	cit[m] <=> icit[m]	288.5075	-43.8291	1000.0000	6.4038	-0.9728	22.1962
AKGDm	akg[m] + nad[m] + coa[m] -> nadh[m] + co2[m] + succoa[m]	300.8971	159.6568	549.2234	6.6788	3.5438	12.1907
CSm	h2o[m] + accoa[m] + oaa[m] -> h[m] + coa[m] + cit[m]	0.0000	0.0000	199.2991	0.0000	0.0000	4.4237
FUMm	h2o[m] + fum[m] <=> mal_L[m]	22.7530	-166.0799	291.2860	0.5050	-3.6863	6.4654
ICDHy	nadp[c] + icit[c] -> nadph[c] + akg[c] + co2[c]	0.0000	0.0000	901.0848	0.0000	0.0000	20.0006
ICDHyp	nadp[x] + icit[x] -> akg[x] + co2[x] + nadph[x]	0.0000	0.0000	901.0848	0.0000	0.0000	20.0006
ICDHyrm	nadp[m] + icit[m] <=> nadph[m] + akg[m] + co2[m]	0.0000	-1000.0000	185.2267	0.0000	-22.1962	4.1113
MDHm	nad[m] + mal_L[m] <=> h[m] + nadh[m] + oaa[m]	-454.5054	-814.0985	564.4821	-10.0883	-18.0699	12.5293
SUCD1m	fad[m] + succ[m] <=> fadh2[m] + fum[m]	300.8971	6.5628	441.6156	6.6788	0.1457	9.8022
SUCOAS1m	coa[m] + gtp[m] + succ[m] <=> pi[m] + gdp[m] + succoa[m]	-300.8971	-549.2234	-159.6568	-6.6788	-12.1907	-3.5438
r0081	akg[m] + ala_L[m] <=> glu_L[m] + pyr[m]	0.0000	0.0000	498.0783	0.0000	0.0000	11.0554
ACONT	cit[c] <=> icit[c]	-288.5075	-500.1230	51.0890	-6.4038	-11.1008	1.1340
FUM	h2o[c] + fum[c] <=> mal_L[c]	522.7416	157.6135	824.4766	11.6029	3.4984	18.3002
MDH	nad[c] + mal_L[c] <=> h[c] + nadh[c] + oaa[c]	1000.0000	-359.2548	1000.0000	22.1962	-7.9741	22.1962

FBA Flux Sum

Pathway	FluxSumPositiveFBA	FluxSumNegativeFBA
Citric acid cycle	7438.6970	-7438.6970

Pyruvate metabolism

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVAmin	Normali-zedFVA-max
ACTNMO	o2[c] + h[c] + nadph[c] + acetone[c] -> h2o[c] + nadp[c] + acetol[c]	62.2225	0.0000	200.5383	1.3811	0.0000	4.4512
ALCD21_D	nad[c] + 12ppd_R[c] -> h[c] + nadh[c] + lald_D[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
ALCD21_L	nad[c] + 12ppd_S[c] -> h[c] + nadh[c] + lald_L[c]	62.2225	0.0000	1000.0000	1.3811	0.0000	22.1962
ALCD22_D	nad[c] + lald_D[c] -> h[c] + nadh[c] + mthgxl[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
ALCD22_L	nad[c] + lald_L[c] -> h[c] + nadh[c] + mthgxl[c]	62.2225	0.0000	1000.0000	1.3811	0.0000	22.1962
ALR2	h[c] + nadph[c] + mthgxl[c] -> nadp[c] + acetol[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
ALR3	h[c] + nadph[c] + acetol[c] -> nadp[c] + 12ppd_S[c]	62.2225	0.0000	1000.0000	1.3811	0.0000	22.1962
GLYOXm	h2o[m] + lgt_S[m] -> h[m] + lac_D[m] + gthrd[m]	65.9900	0.0000	230.0747	1.4647	0.0000	5.1068
LALDO	nad[c] + lald_D[c] + gthrd[c] <=> h[c] + nadh[c] + lgt_S[c]	0.0000	-1000.0000	230.0747	0.0000	-22.1962	5.1068
LCADi_D	h2o[c] + nad[c] + lald_D[c] -> 2 h[c] + nadh[c] + lac_D[c]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651
LCADim	h2o[m] + nad[m] + lald_L[m] -> 2 h[m] + nadh[m] + lac_L[m]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651
LDH_Lm	nad[m] + lac_L[m] <=> h[m] + nadh[m] + pyr[m]	271.4764	0.0000	603.7198	6.0257	0.0000	13.4003
MGSA2	g3p[c] -> pi[c] + mthgxl[c]	0.0000	0.0000	214.7597	0.0000	0.0000	4.7668
PPDOy	h[c] + nadph[c] + lald_D[c] -> nadp[c] + 12ppd_R[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
r0818	dump[m] -> dump[c]	0.0000	0.0000	117.5328	0.0000	0.0000	2.6088
GLYOX	h2o[c] + lgt_S[c] -> h[c] + gthrd[c] + lac_D[c]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651
LCADi	h2o[c] + nad[c] + lald_L[c] -> 2 h[c] + nadh[c] + lac_L[c]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651
LGTHL	mthgxl[c] + gthrd[c] -> lgt_S[c]	65.9900	0.0000	1000.0000	1.4647	0.0000	22.1962
HMR_3855	h[c] + nadph[c] + lald_L[c] -> nadp[c] + 12ppd_R[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
HMR_8501	h[c] + nadph[c] + mthgxl[c] -> nadp[c] + lald_D[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962

FBA Flux Sum

Pathway	FluxSumPositiveFBA	FluxSumNegativeFBA
Pyruvate metabolism	1762.8365	-1491.3601

Glutamate metabolism

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVAmin	Normali-zedFVA-max
ABTArm	akg[m] + 4abut[m] <=> glu_L[m] + sucsal[m]	0.0000	-265.4679	28.4545	0.0000	-5.8924	0.6316
GLUDxm	h2o[m] + nad[m] + glu_L[m] <=> h[m] + akg[m] + nadh[m] + nh4[m]	-490.1987	-1000.0000	998.9961	-10.8805	-22.1962	22.1739
GLUDym	h2o[m] + nadp[m] + glu_L[m] <=> h[m] + nadph[m] + akg[m] + nh4[m]	259.4611	-1000.0000	998.9961	5.7590	-22.1962	22.1739
GLUNm	h2o[m] + gln_L[m] -> glu_L[m] + nh4[m]	230.7376	0.0000	498.0783	5.1215	0.0000	11.0554
GTHO	h[c] + nadph[c] + gthox[c] -> nadp[c] + 2 gthrd[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
GTHOm	h[m] + nadph[m] + gthox[m] -> nadp[m] + 2 gthrd[m]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
P5CDm	2 h2o[m] + nad[m] + 1pyr5c[m] -> h[m] + nadh[m] + glu_L[m]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
r0178	h2o[m] + nad[m] + sucsal[m] <=> 2 h[m] + nadh[m] + succ[m]	0.0000	-265.4679	28.4545	0.0000	-5.8924	0.6316
GLNS	nh4[c] + atp[c] + glu_L[c] -> h[c] + adp[c] + pi[c] + gln_L[c]	0.0000	0.0000	135.1627	0.0000	0.0000	3.0001
HMR_3831	asp_L[c] -> nh4[c] + fum[c]	227.8262	0.0000	648.8795	5.0569	0.0000	14.4026

FBA Flux Sum

Pathway	FluxSumPositiveFBA	FluxSumNegativeFBA
Glutamate metabolism	3425.5679	-3428.4793

Alanine and aspartate metabolism

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	NormalizedF-BA	NormalizedF-VAmin	NormalizedF-VAmax
ASPTAm	akg[m] + asp_L[m] <=> glu_L[m] + oaa[m]	0.0000	-15.9690	318.9484	0.0000	-0.3545	7.0794
ALAR	ala_L[c] <=> ala_D[c]	0.0000	0.0000	607.2141	0.0000	0.0000	13.4778
ASPTA	akg[c] + asp_L[c] <=> glu_L[c] + oaa[c]	-256.9871	-675.1242	-12.5888	-5.7041	-14.9852	-0.2794

FBA Flux Sum

Pathway	FluxSumPositiveFBA	FluxSumNegativeFBA
Alanine and aspartate metabolism	513.9741	-513.9741

Fluxes Per Metabolite

D-Glucose (glc_D)

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVAmin	Normali-zedFVAmax
CBPPer	h[r] + glc_D[r] + cbp[r] -> co2[r] + nh4[r] + g6p[r]	0.0000	0.0000	79.5075	0.0000	0.0000	1.7648
DOLGPP_Ler	h2o[r] + 0.1 dolglcp_L[r] -> h[r] + 0.1 dolp_L[r] + glc_D[r]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
G6PPer	h2o[r] + g6p[r] -> glc_D[r] + pi[r]	0.0000	0.0000	135.1627	0.0000	0.0000	3.0001
GBA	h2o[c] + gluside_hs[c] -> glc_D[c] + crm_hs[c]	0.0000	0.0000	67.5814	0.0000	0.0000	1.5000
GBAl	h2o[l] + gluside_hs[l] -> crm_hs[l] + glc_D[l]	0.0000	0.0000	16.8953	0.0000	0.0000	0.3750
GLCt1r	glc_D[e] <=> glc_D[c]	521.0941	451.2757	972.1950	11.5663	10.0166	21.5790
GLCter	glc_D[c] <=> glc_D[r]	0.0000	-135.1627	79.5075	0.0000	-3.0001	1.7648
GLCtg	glc_D[c] <=> glc_D[g]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
GLCtly	glc_D[l] -> glc_D[c]	0.0000	0.0000	16.8953	0.0000	0.0000	0.3750
GLDBRAN	h2o[c] + dxtrn[c] -> glc_D[c] + glygn3[c]	0.0000	0.0000	7.1138	0.0000	0.0000	0.1579
GLPASE2	7 h2o[c] + glygn3[c] -> 7 glc_D[c] + Tyr_ggn[c]	0.0000	0.0000	7.1138	0.0000	0.0000	0.1579
LACZe	h2o[e] + lcts[e] -> glc_D[e] + gal[e]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
SBTR	h[c] + nadph[c] + glc_D[c] -> nadp[c] + sbt_D[c]	0.0000	0.0000	974.3168	0.0000	0.0000	21.6261
UGALGTg	udpgal[g] + glc_D[g] -> h[g] + udp[g] + lcts[g]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
r0354	itp[c] + glc_D[c] -> h[c] + idp[c] + g6p[c]	0.0000	0.0000	974.3168	0.0000	0.0000	21.6261
r0355	datp[c] + glc_D[c] -> h[c] + dadp[c] + g6p[c]	0.0000	0.0000	974.3168	0.0000	0.0000	21.6261
HEX1	atp[c] + glc_D[c] -> h[c] + adp[c] + g6p[c]	521.0941	0.0000	974.3168	11.5663	0.0000	21.6261
GBA2e	h2o[e] + gluside_hs[e] -> glc_D[e] + crm_hs[e]	0.0000	0.0000	27.0325	0.0000	0.0000	0.6000
EX_glc_D[e]	glc_D[e] <=>	-521.0941	-972.1950	-451.2757	-11.5663	-21.5790	-10.0166

FBA Flux Sum

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
glc_D[e]	521.0941	-521.0941
glc_D[c]	521.0941	-521.0941

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
glc_D[r]	0.0000	0.0000
glc_D[l]	0.0000	0.0000
glc_D[g]	0.0000	0.0000

Pyruvate (pyr)

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVAmin	Normali-zedFVA-max
3SPYRSP	h2o[c] + 3snpyr[c] -> h[c] + pyr[c] + so3[c]	0.0000	0.0000	30.6511	0.0000	0.0000	0.6803
3SPYRSPm	h2o[m] + 3snpyr[m] -> h[m] + pyr[m] + so3[m]	0.0000	0.0000	15.9690	0.0000	0.0000	0.3545
ACNMLr	acnam[c] <=> pyr[c] + acmana[c]	0.0000	0.0000	67.5814	0.0000	0.0000	1.5000
DDPGAm	4h2oglt[m] <=> pyr[m] + glx[m]	0.0000	0.0000	318.9484	0.0000	0.0000	7.0794
LDH_Lm	nad[m] + lac_L[m] <=> h[m] + nadh[m] + pyr[m]	271.4764	0.0000	603.7198	6.0257	0.0000	13.4003
PDHm	nad[m] + coa[m] + pyr[m] -> nadh[m] + co2[m] + accoa[m]	271.4764	0.0000	603.7198	6.0257	0.0000	13.4003
PYRt2p	h[c] + pyr[c] <=> h[x] + pyr[x]	0.0000	-477.5753	180.2170	0.0000	-10.6003	4.0001
r0081	akg[m] + ala_L[m] <=> glu_L[m] + pyr[m]	0.0000	0.0000	498.0783	0.0000	0.0000	11.0554
r0156	pyr[c] + gln_L[c] -> ala_L[c] + HC00591[c]	288.5075	69.5483	652.8496	6.4038	1.5437	14.4908
r0157	pyr[m] + gln_L[m] -> ala_L[m] + HC00591[m]	0.0000	0.0000	498.0783	0.0000	0.0000	11.0554
r0173	nad[x] + lac_L[x] <=> h[x] + nadh[x] + pyr[x]	0.0000	-180.2170	477.5753	0.0000	-4.0001	10.6003
r0193	h2o[c] + cys_L[c] <=> nh4[c] + h[c] + pyr[c] + HC00250[c]	0.0000	-16.0193	0.0000	0.0000	-0.3556	0.0000
r0595	h[c] + so3[c] + mercppyr[c] -> pyr[c] + HC01501[c]	0.0000	0.0000	16.0193	0.0000	0.0000	0.3556
RE2954C	h[c] + pep[c] + dtdp[c] <=> pyr[c] + dttp[c]	288.5075	159.6568	1000.0000	6.4038	3.5438	22.1962
LDH_L	nad[c] + lac_L[c] <=> h[c] + pyr[c] + nadh[c]	-1000.0000	-1000.0000	-511.1446	-22.1962	-22.1962	-11.3454
PYK	h[c] + adp[c] + pep[c] -> atp[c] + pyr[c]	1000.0000	159.6568	1000.0000	22.1962	3.5438	22.1962
HMR_6515	so3[c] + mercppyr[c] -> pyr[c] + tsul[c]	0.0000	0.0000	30.6511	0.0000	0.0000	0.6803

FBA Flux Sum

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
pyr[c]	1288.5075	-1288.5075
pyr[m]	271.4764	-271.4764
pyr[x]	0.0000	0.0000

(S)-Lactate (lac_L)

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVAmin	Normali-zedFVAmax
LCADim	h2o[m] + nad[m] + lald_L[m] -> 2 h[m] + nadh[m] + lac_L[m]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651
LDH_Lm	nad[m] + lac_L[m] <=> h[m] + nadh[m] + pyr[m]	271.4764	0.0000	603.7198	6.0257	0.0000	13.4003
L_LACtcm	lac_L[c] -> lac_L[m]	271.4764	0.0000	603.7198	6.0257	0.0000	13.4003
r0173	nad[x] + lac_L[x] <=> h[x] + nadh[x] + pyr[x]	0.0000	-180.2170	477.5753	0.0000	-4.0001	10.6003
r2516	h[x] + lac_L[x] <=> h[c] + lac_L[c]	0.0000	-1000.0000	180.2170	0.0000	-22.1962	4.0001
LACLt	lac_L[x] -> lac_L[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
EX_lac_L[e]	lac_L[e] ->	728.5236	89.3018	1000.0000	16.1704	1.9822	22.1962
L_LACt2r	h[e] + lac_L[e] <=> h[c] + lac_L[c]	-728.5236	-1000.0000	-89.3018	-16.1704	-22.1962	-1.9822
LCADi	h2o[c] + nad[c] + lald_L[c] -> 2 h[c] + nadh[c] + lac_L[c]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651
LDH_L	nad[c] + lac_L[c] <=> h[c] + pyr[c] + nadh[c]	-1000.0000	-1000.0000	-511.1446	-22.1962	-22.1962	-11.3454

FBA Flux Sum

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
lac_L[m]	271.4764	-271.4764
lac_L[c]	1000.0000	-1000.0000
lac_L[x]	0.0000	0.0000
lac_L[e]	728.5236	-728.5236

(R)-Lactate (lac_D)

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVAmin	Normali-zedFVAmax
D_LACt2	h[e] + lac_D[e] <=> h[c] + lac_D[c]	-65.9900	-230.0747	0.0000	-1.4647	-5.1068	0.0000
D_LACtm	h[c] + lac_D[c] <=> h[m] + lac_D[m]	-65.9900	-230.0747	0.0000	-1.4647	-5.1068	0.0000
EX_lac_D[e]	lac_D[e] ->	65.9900	0.0000	230.0747	1.4647	0.0000	5.1068
GLYOXm	h2o[m] + lgt_S[m] -> h[m] + lac_D[m] + gthrd[m]	65.9900	0.0000	230.0747	1.4647	0.0000	5.1068
LCADi_D	h2o[c] + nad[c] + lald_D[c] -> 2 h[c] + nadh[c] + lac_D[c]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651
GLYOX	h2o[c] + lgt_S[c] -> h[c] + gthrd[c] + lac_D[c]	0.0000	0.0000	196.6606	0.0000	0.0000	4.3651

FBA Flux Sum

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
lac_D[e]	65.9900	-65.9900
lac_D[c]	65.9900	-65.9900
lac_D[m]	65.9900	-65.9900

L-Glutamine (gln_L)

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	NormalizedFBA	NormalizedFVA-min	NormalizedFVA-max
GLNtm	gln_L[c] -> gln_L[m]	230.7376	1.0039	498.0783	5.1215	0.0223	11.0554
GLNtN1	h[c] + 2 na1[e] + gln_L[e] <=> h[e] + 2 na1[c] + gln_L[c]	549.2802	187.6922	978.4781	12.1919	4.1660	21.7185
GLUNm	h2o[m] + gln_L[m] -> glu_L[m] + nh4[m]	230.7376	0.0000	498.0783	5.1215	0.0000	11.0554
r0156	pyr[c] + gln_L[c] -> ala_L[c] + HC00591[c]	288.5075	69.5483	652.8496	6.4038	1.5437	14.4908
r0157	pyr[m] + gln_L[m] -> ala_L[m] + HC00591[m]	0.0000	0.0000	498.0783	0.0000	0.0000	11.0554
biomass_reaction	20.6508 h2o[c] + 20.7045 atp[c] + 0.385872 glu_L[c] + 0.352607 asp_L[c] + 0.036117 gtp[c] + 0.505626 ala_L[c] + 0.279425 asn_L[c] + 0.046571 cys_L[c] + 0.325996 gln_L[c] + 0.538891 gly[c] + 0.392525 ser_L[c] + 0.31269 thr_L[c] + 0.592114 lys_L[c] + 0.35926 arg_L[c] + 0.153018 met_L[c] + 0.023315 pail_hs[c] + 0.039036 ctp[c] + 0.154463 pchol_hs[c] + 0.055374 pe_hs[c] + 0.020401 chsterol[c] + 0.002914 pglyc_hs[c] + 0.011658 clpn_hs[c] + 0.009898 dgtp[n] + 0.009442 dctp[n] + 0.013183 datp[n] + 0.053446 utp[c] + 0.013091 dttp[n] + 0.275194 g6p[c] + 0.126406 his_L[c] + 0.159671 tyr_L[c] + 0.286078 ile_L[c] + 0.545544 leu_L[c] + 0.013306 trp_L[c] + 0.259466 phe_L[c] + 0.412484 pro_L[c] + 0.005829 ps_hs[c] + 0.017486 sphmyln_hs[c] + 0.352607 val_L[c] -> 20.6508 h[c] + 20.6508 adp[c] + 20.6508 pi[c]	45.0528	40.5475	45.0528	1.0000	0.9000	1.0000
biomass_maintenance	20.6508 h2o[c] + 20.7045 atp[c] + 0.38587 glu_L[c] + 0.35261 asp_L[c] + 0.036117 gtp[c] + 0.50563 ala_L[c] + 0.27942 asn_L[c] + 0.046571 cys_L[c] + 0.326 gln_L[c] + 0.53889 gly[c] + 0.39253 ser_L[c] + 0.31269 thr_L[c] + 0.59211 lys_L[c] + 0.35926 arg_L[c] + 0.15302 met_L[c] + 0.023315 pail_hs[c] + 0.039036 ctp[c] + 0.15446 pchol_hs[c] + 0.055374 pe_hs[c] + 0.020401 chsterol[c] + 0.002914 pglyc_hs[c] + 0.011658 clpn_hs[c] + 0.053446 utp[c] + 0.27519 g6p[c] + 0.12641 his_L[c] + 0.15967 tyr_L[c] + 0.28608 ile_L[c] + 0.54554 leu_L[c] + 0.013306 trp_L[c] + 0.25947 phe_L[c] + 0.41248 pro_L[c] + 0.005829 ps_hs[c] + 0.017486 sphmyln_hs[c] + 0.35261 val_L[c] -> 20.6508 h[c] + 20.6508 adp[c] + 20.6508 pi[c]	0.0000	0.0000	4.7134	0.0000	0.0000	0.1046
CBPS	h2o[c] + 2 atp[c] + hco3[c] + gln_L[c] -> 2 h[c] + 2 adp[c] + pi[c] + glu_L[c] + cbp[c]	5.1818	4.6636	84.1711	0.1150	0.1035	1.8683

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVA-min	Normal-izedFVA-max
CTPS2	h2o[c] + atp[c] + gln_L[c] + utp[c] -> 2 h[c] + adp[c] + pi[c] + glu_L[c] + ctp[c]	0.0000	0.0000	113.5937	0.0000	0.0000	2.5213
GF6PTA	gln_L[c] + f6p[c] -> glu_L[c] + gam6p[c]	0.0000	0.0000	568.5009	0.0000	0.0000	12.6185
GLNS	nh4[c] + atp[c] + glu_L[c] -> h[c] + adp[c] + pi[c] + gln_L[c]	0.0000	0.0000	135.1627	0.0000	0.0000	3.0001
GMPS2	h2o[c] + atp[c] + gln_L[c] + xmp[c] -> 2 h[c] + glu_L[c] + amp[c] + ppi[c] + gmp[c]	0.0000	0.0000	61.7988	0.0000	0.0000	1.3717
EX_gln_L[e]	gln_L[e] <=>	-549.2802	-978.4781	-187.6922	-12.1919	-21.7185	-4.1660
GLUPRT	h2o[c] + gln_L[c] + prpp[c] -> glu_L[c] + ppi[c] + pram[c]	5.0831	4.5748	12.9610	0.1128	0.1015	0.2877
PRFGS	h2o[c] + atp[c] + gln_L[c] + fgam[c] -> h[c] + adp[c] + pi[c] + glu_L[c] + fpram[c]	5.0831	4.5748	12.9610	0.1128	0.1015	0.2877

FBA Flux Sum

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
gln_L[c]	549.2802	-549.2802
gln_L[e]	549.2802	-549.2802
gln_L[m]	230.7376	-230.7376

L-Glutamate (glu_L)

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVA-min	Normal-izedFVA-max
3SALATAi	h[c] + akg[c] + 3sala[c] -> glu_L[c] + 3snpyr[c]	0.0000	0.0000	30.6511	0.0000	0.0000	0.6803
3SALATAim	h[m] + akg[m] + 3sala[m] -> glu_L[m] + 3snpyr[m]	0.0000	0.0000	15.9690	0.0000	0.0000	0.3545
ABTArm	akg[m] + 4abut[m] <=> glu_L[m] + sucsal[m]	0.0000	-265.4679	28.4545	0.0000	-5.8924	0.6316
ASPTAm	akg[m] + asp_L[m] <=> glu_L[m] + oaa[m]	0.0000	-15.9690	318.9484	0.0000	-0.3545	7.0794
CYSTA	akg[c] + cys_L[c] <=> glu_L[c] + mercppyr[c]	0.0000	0.0000	46.5484	0.0000	0.0000	1.0332
EHGLATm	akg[m] + e4hglu[m] -> glu_L[m] + 4h2oglt[m]	0.0000	0.0000	318.9484	0.0000	0.0000	7.0794
FPGS2	atp[c] + glu_L[c] + 5thf[c] -> h[c] + adp[c] + pi[c] + 6thf[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
FPGS2m	glu_L[m] + 5thf[m] + atp[m] -> h[m] + 6thf[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
FPGS3	atp[c] + glu_L[c] + 6thf[c] -> h[c] + adp[c] + pi[c] + 7thf[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
FPGS3m	glu_L[m] + 6thf[m] + atp[m] -> h[m] + 7thf[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
FPGS4	atp[c] + 4 glu_L[c] + dhf[c] -> 3 h2o[c] + h[c] + adp[c] + pi[c] + 5dhf[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
FPGS4m	4 glu_L[m] + atp[m] + dhf[m] -> 3 h2o[m] + h[m] + adp[m] + pi[m] + 5dhf[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
FPGS5	atp[c] + glu_L[c] + 5dhf[c] -> h[c] + adp[c] + pi[c] + 6dhf[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
FPGS5m	glu_L[m] + atp[m] + 5dhf[m] -> h[m] + 6dhf[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
FPGS6	atp[c] + glu_L[c] + 6dhf[c] -> h[c] + adp[c] + pi[c] + 7dhf[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
FPGS6m	glu_L[m] + 6dhf[m] + atp[m] -> h[m] + 7dhf[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
FPGS7m	10fthf[m] + 4 glu_L[m] + atp[m] -> 10fthf5glu[m] + 3 h2o[m] + h[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
FPGS8	10fthf5glu[c] + atp[c] + glu_L[c] -> 10fthf6glu[c] + h[c] + adp[c] + pi[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
FPGS8m	10fthf5glu[m] + glu_L[m] + atp[m] -> 10fthf6glu[m] + h[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVA-min	Normal-izedFVA-max
FPGS9	10fthf6glu[c] + atp[c] + glu_L[c] -> 10fthf7glu[c] + h[c] + adp[c] + pi[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
FPGS9m	10fthf6glu[m] + glu_L[m] + atp[m] -> 10fthf7glu[m] + h[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
FPGSm	4 glu_L[m] + atp[m] + thf[m] -> 3 h2o[m] + h[m] + 5thf[m] + adp[m] + pi[m]	0.0000	0.0000	45.6752	0.0000	0.0000	1.0138
GGH_10FTHF5GLUl	10fthf5glu[l] + 4 h2o[l] -> 10fthf[l] + 4 glu_L[l]	0.0000	0.0000	47.2792	0.0000	0.0000	1.0494
GGH_10FTHF6GLUl	10fthf6glu[l] + h2o[l] -> 10fthf5glu[l] + glu_L[l]	0.0000	0.0000	47.2792	0.0000	0.0000	1.0494
GGH_10FTHF7GLUl	10fthf7glu[l] + h2o[l] -> 10fthf6glu[l] + glu_L[l]	0.0000	0.0000	47.2792	0.0000	0.0000	1.0494
GGH_5DHFl	4 h2o[l] + 5dhf[l] -> dhf[l] + 4 glu_L[l]	0.0000	0.0000	47.2792	0.0000	0.0000	1.0494
GGH_6DHFl	h2o[l] + 6dhf[l] -> 5dhf[l] + glu_L[l]	0.0000	0.0000	47.2792	0.0000	0.0000	1.0494
GGH_7DHFl	h2o[l] + 7dhf[l] -> 6dhf[l] + glu_L[l]	0.0000	0.0000	47.2792	0.0000	0.0000	1.0494
GGT6	glu_L[c] + leuktrE4[c] <=> h2o[c] + leuktrF4[c]	0.0000	-1000.0000	1000.0000	0.0000	-22.1962	22.1962
GLU5Km	glu_L[m] + atp[m] -> adp[m] + glu5p[m]	0.0000	0.0000	327.2199	0.0000	0.0000	7.2630
GLUCYS	atp[c] + glu_L[c] + cys_L[c] -> h[c] + adp[c] + pi[c] + glucys[c]	0.0000	0.0000	135.1627	0.0000	0.0000	3.0001
GLUDxm	h2o[m] + nad[m] + glu_L[m] <=> h[m] + akg[m] + nadh[m] + nh4[m]	-490.1987	-1000.0000	998.9961	-10.8805	-22.1962	22.1739
GLUDym	h2o[m] + nadp[m] + glu_L[m] <=> h[m] + nadph[m] + akg[m] + nh4[m]	259.4611	-1000.0000	998.9961	5.7590	-22.1962	22.1739
GLUNm	h2o[m] + gln_L[m] -> glu_L[m] + nh4[m]	230.7376	0.0000	498.0783	5.1215	0.0000	11.0554
GLUt7l	glu_L[l] -> glu_L[c]	0.0000	0.0000	283.6751	0.0000	0.0000	6.2965
ILETAm	akg[m] + ile_L[m] <=> glu_L[m] + 3mop[m]	0.0000	-1000.0000	0.0000	0.0000	-22.1962	0.0000
LCYSTAT	Lcyst[c] + akg[c] <=> glu_L[c] + 3spyr[c]	0.0000	0.0000	30.6511	0.0000	0.0000	0.6803
LEUTAm	akg[m] + leu_L[m] <=> glu_L[m] + 4mop[m]	0.0000	-1000.0000	0.0000	0.0000	-22.1962	0.0000
ORNTArm	akg[m] + orn[m] <=> glu_L[m] + glu5sa[m]	0.0000	-209.9039	9.4137	0.0000	-4.6591	0.2089
P5CDm	2 h2o[m] + nad[m] + 1pyr5c[m] -> h[m] + nadh[m] + glu_L[m]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
PHETA1m	akg[m] + phe_L[m] <=> glu_L[m] + phpyr[m]	0.0000	0.0000	901.0848	0.0000	0.0000	20.0006
SACCD4m	h2o[m] + nadp[m] + saccrp_L[m] -> h[m] + nadph[m] + glu_L[m] + L2aadp6sa[m]	0.0000	0.0000	187.6944	0.0000	0.0000	4.1661

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVA-min	Normal-izedFVA-max
TYRTAm	akg[m] + tyr_L[m] <=> 34hpp[m] + glu_L[m]	5.1818	4.6636	7.8971	0.1150	0.1035	0.1753
VALTAm	akg[m] + val_L[m] <=> glu_L[m] + 3mob[m]	-448.0734	-1000.0000	0.0000	-9.9455	-22.1962	0.0000
r0074	h2o[m] + nad[m] + glu5sa[m] <=> 2 h[m] + nadh[m] + glu_L[m]	-18.5836	-1000.0000	310.4947	-0.4125	-22.1962	6.8918
r0081	akg[m] + ala_L[m] <=> glu_L[m] + pyr[m]	0.0000	0.0000	498.0783	0.0000	0.0000	11.0554
r0129	h2o[c] + gthrd[c] <=> glu_L[c] + cgly[c]	-197.7279	-743.0106	27.0347	-4.3888	-16.4920	0.6001
r0130	h2o[e] + gthrd[e] <=> cgly[e] + glu_L[e]	230.2673	1.0749	774.6297	5.1111	0.0239	17.1938
r0450	L2aadp[m] + akg[m] <=> 2oxoadp[m] + glu_L[m]	0.0000	0.0000	187.6944	0.0000	0.0000	4.1661
r0641	h2o[c] + leuktrC4[c] <=> glu_L[c] + leuktrD4[c]	0.0000	-1000.0000	1000.0000	0.0000	-22.1962	22.1962
r1382	6 atp[c] + 6 glu_L[c] + thf[c] -> 6 h[c] + 6 adp[c] + 6 pi[c] + 7thf[c]	0.0000	0.0000	22.5271	0.0000	0.0000	0.5000
r1383	6 h2o[c] + 7thf[c] -> 6 glu_L[c] + thf[c]	0.0000	0.0000	47.2792	0.0000	0.0000	1.0494
biomass_reaction	20.6508 h2o[c] + 20.7045 atp[c] + 0.385872 glu_L[c] + 0.352607 asp_L[c] + 0.036117 gtp[c] + 0.505626 ala_L[c] + 0.279425 asn_L[c] + 0.046571 cys_L[c] + 0.325996 gln_L[c] + 0.538891 gly[c] + 0.392525 ser_L[c] + 0.31269 thr_L[c] + 0.592114 lys_L[c] + 0.35926 arg_L[c] + 0.153018 met_L[c] + 0.023315 pail_hs[c] + 0.039036 ctp[c] + 0.154463 pchol_hs[c] + 0.055374 pe_hs[c] + 0.020401 chsterol[c] + 0.002914 pglyc_hs[c] + 0.011658 clpn_hs[c] + 0.009898 dgtp[n] + 0.009442 dctp[n] + 0.013183 datp[n] + 0.053446 utp[c] + 0.013091 dttp[n] + 0.275194 g6p[c] + 0.126406 his_L[c] + 0.159671 tyr_L[c] + 0.286078 ile_L[c] + 0.545544 leu_L[c] + 0.013306 trp_L[c] + 0.259466 phe_L[c] + 0.412484 pro_L[c] + 0.005829 ps_hs[c] + 0.017486 sphmyln_hs[c] + 0.352607 val_L[c] -> 20.6508 h[c] + 20.6508 adp[c] + 20.6508 pi[c]	45.0528	40.5475	45.0528	1.0000	0.9000	1.0000
biomass_maintenance	20.6508 h2o[c] + 20.7045 atp[c] + 0.38587 glu_L[c] + 0.35261 asp_L[c] + 0.036117 gtp[c] + 0.50563 ala_L[c] + 0.27942 asn_L[c] + 0.046571 cys_L[c] + 0.326 gln_L[c] + 0.53889 gly[c] + 0.39253 ser_L[c] + 0.31269 thr_L[c] + 0.59211 lys_L[c] + 0.35926 arg_L[c] + 0.15302 met_L[c] + 0.023315	0.0000	0.0000	4.7134	0.0000	0.0000	0.1046

ReactionID	Formula	FBA	FVAmin	FVAmax	Normali-zedFBA	Normali-zedFVA-min	Normal-izedFVA-max
	pail_hs[c] + 0.039036 ctp[c] + 0.15446 pchol_hs[c] + 0.055374 pe_hs[c] + 0.020401 chsterol[c] + 0.002914 pglyc_hs[c] + 0.011658 clpn_hs[c] + 0.053446 utp[c] + 0.27519 g6p[c] + 0.12641 his_L[c] + 0.15967 tyr_L[c] + 0.28608 ile_L[c] + 0.54554 leu_L[c] + 0.013306 trp_L[c] + 0.25947 phe_L[c] + 0.41248 pro_L[c] + 0.005829 ps_hs[c] + 0.017486 sphmyln_hs[c] + 0.35261 val_L[c] -> 20.6508 h[c] + 20.6508 adp[c] + 20.6508 pi[c]						
UNK3r	glu_L[c] + 2kmb[c] <=> akg[c] + met_L[c]	0.0000	0.0000	31.0719	0.0000	0.0000	0.6897
ASPTA	akg[c] + asp_L[c] <=> glu_L[c] + oaa[c]	-256.9871	-675.1242	-12.5888	-5.7041	-14.9852	-0.2794
CBPS	h2o[c] + 2 atp[c] + hco3[c] + gln_L[c] -> 2 h[c] + 2 adp[c] + pi[c] + glu_L[c] + cbp[c]	5.1818	4.6636	84.1711	0.1150	0.1035	1.8683
CTPS2	h2o[c] + atp[c] + gln_L[c] + utp[c] -> 2 h[c] + adp[c] + pi[c] + glu_L[c] + ctp[c]	0.0000	0.0000	113.5937	0.0000	0.0000	2.5213
EX_glu_L[e]	glu_L[e] <=>	230.2673	1.0749	774.6297	5.1111	0.0239	17.1938
FPGS7	10fthf[c] + atp[c] + 4 glu_L[c] -> 10fthf5glu[c] + 3 h2o[c] + h[c] + adp[c] + pi[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
GF6PTA	gln_L[c] + f6p[c] -> glu_L[c] + gam6p[c]	0.0000	0.0000	568.5009	0.0000	0.0000	12.6185
GLNS	nh4[c] + atp[c] + glu_L[c] -> h[c] + adp[c] + pi[c] + gln_L[c]	0.0000	0.0000	135.1627	0.0000	0.0000	3.0001
GMPS2	h2o[c] + atp[c] + gln_L[c] + xmp[c] -> 2 h[c] + glu_L[c] + amp[c] + ppi[c] + gmp[c]	0.0000	0.0000	61.7988	0.0000	0.0000	1.3717
ILETA	akg[c] + ile_L[c] <=> glu_L[c] + 3mop[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
LEUTA	akg[c] + leu_L[c] <=> glu_L[c] + 4mop[c]	0.0000	0.0000	1000.0000	0.0000	0.0000	22.1962
VALTA	akg[c] + val_L[c] <=> glu_L[c] + 3mob[c]	448.0734	0.0000	1000.0000	9.9455	0.0000	22.1962
FPGS	atp[c] + 4 glu_L[c] + thf[c] -> 3 h2o[c] + h[c] + adp[c] + pi[c] + 5thf[c]	0.0000	0.0000	45.0542	0.0000	0.0000	1.0000
GLUPRT	h2o[c] + gln_L[c] + prpp[c] -> glu_L[c] + ppi[c] + pram[c]	5.0831	4.5748	12.9610	0.1128	0.1015	0.2877
PHETA1	akg[c] + phe_L[c] <=> glu_L[c] + phpyr[c]	0.0000	-901.0848	475.2343	0.0000	-20.0006	10.5484
PRFGS	h2o[c] + atp[c] + gln_L[c] + fgam[c] -> h[c] + adp[c] + pi[c] + glu_L[c] + fpram[c]	5.0831	4.5748	12.9610	0.1128	0.1015	0.2877
TYRTA	akg[c] + tyr_L[c] <=> 34hpp[c] + glu_L[c]	8.6782	0.0000	173.4474	0.1926	0.0000	3.8499

FBA Flux Sum

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
glu_L[m]	726.1180	-726.1180
glu_L[c]	472.0996	-472.0996
glu_L[e]	230.2673	-230.2673
glu_L[l]	0.0000	0.0000

L-Alanine (ala_L)

Fluxes

ReactionID	Formula	FBA	FVAmin	FVAmax	Normal-izedFBA	Normali-zedFVA-min	Normali-zedFVA-max
LFORKYNHYD	h2o[c] + Lfmkynr[c] -> h[c] + ala_L[c] + nformanth[c]	0.0000	0.0000	244.2825	0.0000	0.0000	5.4221
r0081	akg[m] + ala_L[m] <=> glu_L[m] + pyr[m]	0.0000	0.0000	498.0783	0.0000	0.0000	11.0554
r0156	pyr[c] + gln_L[c] -> ala_L[c] + HC00591[c]	288.5075	69.5483	652.8496	6.4038	1.5437	14.4908
r0157	pyr[m] + gln_L[m] -> ala_L[m] + HC00591[m]	0.0000	0.0000	498.0783	0.0000	0.0000	11.0554
r1563	ala_L[e] + tyr_L[c] <=> ala_L[c] + tyr_L[e]	-21.0535	-614.7486	0.0000	-0.4673	-13.6451	0.0000
biomass_reaction	20.6508 h2o[c] + 20.7045 atp[c] + 0.385872 glu_L[c] + 0.352607 asp_L[c] + 0.036117 gtp[c] + 0.505626 ala_L[c] + 0.279425 asn_L[c] + 0.046571 cys_L[c] + 0.325996 gln_L[c] + 0.538891 gly[c] + 0.392525 ser_L[c] + 0.31269 thr_L[c] + 0.592114 lys_L[c] + 0.35926 arg_L[c] + 0.153018 met_L[c] + 0.023315 pail_hs[c] + 0.039036 ctp[c] + 0.154463 pchol_hs[c] + 0.055374 pe_hs[c] + 0.020401 chsterol[c] + 0.002914 pglyc_hs[c] + 0.011658 clpn_hs[c] + 0.009898 dgtp[n] + 0.009442 dctp[n] + 0.013183 datp[n] + 0.053446 utp[c] + 0.013091 dttp[n] + 0.275194 g6p[c] + 0.126406 his_L[c] + 0.159671 tyr_L[c] + 0.286078 ile_L[c] + 0.545544 leu_L[c] + 0.013306 trp_L[c] + 0.259466 phe_L[c] + 0.412484 pro_L[c] + 0.005829 ps_hs[c] + 0.017486 sphmyln_hs[c] + 0.352607 val_L[c] -> 20.6508 h[c] + 20.6508 adp[c] + 20.6508 pi[c]	45.0528	40.5475	45.0528	1.0000	0.9000	1.0000
ALAALACNc	h2o[c] + alaala[c] <=> 2 ala_L[c]	-122.3371	-320.0266	0.0000	-2.7154	-7.1034	0.0000
4HPROLTASCT1	na1[e] + ala_L[e] + 4hpro_LT[c] -> na1[c] + ala_L[c] + 4hpro_LT[e]	0.0000	0.0000	167.8775	0.0000	0.0000	3.7262
biomass_maintenance	20.6508 h2o[c] + 20.7045 atp[c] + 0.38587 glu_L[c] + 0.35261 asp_L[c] + 0.036117 gtp[c] + 0.50563 ala_L[c] + 0.27942 asn_L[c] + 0.046571 cys_L[c] + 0.326 gln_L[c] + 0.53889 gly[c] + 0.39253 ser_L[c] + 0.31269 thr_L[c] + 0.59211 lys_L[c] + 0.35926 arg_L[c] + 0.15302 met_L[c] + 0.023315 pail_hs[c] + 0.039036 ctp[c] + 0.15446 pchol_hs[c] + 0.055374 pe_hs[c] + 0.020401 chsterol[c] + 0.002914 pglyc_hs[c] + 0.011658 clpn_hs[c] + 0.053446 utp[c] + 0.27519 g6p[c] + 0.12641 his_L[c] + 0.15967 tyr_L[c] + 0.28608 ile_L[c] + 0.54554 leu_L[c] + 0.013306 trp_L[c] + 0.25947 phe_L[c] + 0.41248 pro_L[c] + 0.005829 ps_hs[c] + 0.017486 sphmyln_hs[c] + 0.35261 val_L[c] -> 20.6508 h[c] + 20.6508 adp[c] + 20.6508 pi[c]	0.0000	0.0000	4.7134	0.0000	0.0000	0.1046

ReactionID	Formula	FBA	FVAmin	FVAmax	NormalizedFBA	NormalizedFVAmin	NormalizedFVAmax
ALAR	ala_L[c] <=> ala_D[c]	0.0000	0.0000	607.2141	0.0000	0.0000	13.4778
EX_ala_L[e]	ala_L[e] ->	21.0535	0.0000	607.2141	0.4673	0.0000	13.4778

FBA Flux Sum

Metabolite	FluxSumPositiveFBA	FluxSumNegativeFBA
ala_L[m]	0.0000	0.0000
ala_L[e]	21.0535	-21.0535
ala_L[c]	288.5075	-288.5075